Physics-Informed Neural Networks for Parameter Estimation and Hidden-State Recovery in an SEIRD Epidemic Model with Time-Varying Transmission

Group Project: Mathematical Modelling | Group Size: 5 Students

Member 1 (Lead): Motivation, Formulation, Integration, Notebook & Submissions

Member 2: Mathematical Analysis (R0, Next-Gen Matrix, Stability, Final-Size, Sensitivity)

Member 3: Numerical Solvers (Euler, RK4, DOP853 Verification) | Member 4: PINN Architecture & Synthetic Validation

Member 5: Real-World Outbreak Application, Baselines, Quantitative Metrics & Limitations

ABSTRACT

Compartmental epidemic models are fundamental for understanding disease dynamics, yet standard formulations rely on static transmission rates (beta) that fail to capture human behavioral changes, non-pharmaceutical interventions, and seasonality. Furthermore, epidemiological surveillance routinely captures only manifest outcomes (diagnosed cases and deaths), leaving the susceptible pool and latent incubating populations unobserved. In this project, we develop and systematically validate a Physics-Informed Neural Network (PINN) framework to reconstruct latent states and discover time-varying transmission functions beta(t) in an SEIRD dynamical system. We first derive the basic reproduction number R0 = beta/(gamma + mu) via the next-generation matrix, establish disease-free equilibrium stability, and verify theoretical final-size relations against high-precision numerical simulations. Vectorized forward Euler and fourth-order Runge-Kutta solvers are implemented from scratch and rigorously validated against a DOP853 benchmark (rtol=1e-12, atol=1e-14), demonstrating empirical convergence slopes of 0.999 and 3.694. Across synthetic experiments, our dual-network PINN recovers continuous step-like and sinusoidal transmission profiles under Gaussian noise (RMSE < 0.15) and reconstructs hidden susceptible and exposed compartments from case and death counts alone (RMSE <= 8.8e-3). Finally, we evaluate the PINN on a 120-day outbreak surveillance dataset from Ontario, Canada, using a strict 75%/25% chronological train/test split. Comparing against constant-beta and piecewise-constant classical least-squares baselines, we report an honest evaluation of out-of-sample forward extrapolation, highlighting both the latent state discovery power of PINNs and their extrapolation vulnerabilities outside training regimes.

1. Introduction and Motivation

Infectious disease dynamics represent complex, multiscale biological systems governed by nonlinear contact processes and host-pathogen interactions. Classical epidemiological modeling, rooted in ordinary differential equations (ODEs), traditionally treats the effective transmission rate beta as an invariant physical constant. However, in real-world outbreaks, transmission rates are intrinsically dynamic: they change abruptly in response to public health interventions (such as lockdowns, contact tracing, and mask mandates), vary smoothly due to seasonal climatic forcing, and fluctuate with spontaneous community behavioral adaptations.

Moreover, surveillance data collected during an epidemic are fundamentally incomplete. While public health authorities track confirmed positive diagnoses and disease-induced mortalities with relative accuracy, the underlying numbers of individuals in the incubating (exposed) state and the remaining susceptible population are latent and unobserved. Recently, Scientific Machine Learning (SciML) and Physics-Informed Neural Networks (PINNs) have emerged as a powerful paradigm for integrating empirical observational data directly with physical and biological conservation laws. By embedding governing differential equations into the loss functional through automatic differentiation, PINNs regularize the search space of neural networks, allowing for data-driven parameter discovery and latent state reconstruction without requiring specialized discretization meshes.

2. Project Objectives

1. Formulate a normalized SEIRD compartmental model incorporating time-varying transmission rate beta(t) and document all governing epidemiological assumptions.
2. Conduct a rigorous mathematical analysis of the dynamical system, including derivation of the basic reproduction number R0 via the next-generation matrix, stability conditions for disease-free equilibria, final-size transcendental equations, and global parametric sensitivity analysis.
3. Implement custom vectorized numerical integrators (Forward Euler and Runge-Kutta 4) and verify their empirical convergence rates and conservation invariant preservation against a high-precision DOP853 benchmark solver.

4. Design and validate a dual-network PINN architecture on synthetic datasets to quantify forward integration accuracy, inverse recovery of unknown beta(t) profiles under noise, and latent state reconstruction of unobserved compartments.
5. Apply the proposed methodology to a real-world 120-day epidemiological outbreak dataset (Ontario, Canada) using a strictly segregated 75% training and 25% held-out forecast window, comparing quantitative performance metrics (MAE, RMSE, and Relative Error) against classical least-squares baselines.

3. Model Assumptions

- Closed Population: The total population size N is constant over the observation window. Demographic births and natural deaths balance on the acute outbreak timescale (~100 days).
- Homogeneous Mixing: Every individual has an equal probability of contacting any other individual in the population.
- Permanent Immunity: Recovered individuals (R) acquire lasting sterilizing immunity during the modeled epoch.
- Fixed Biological Timescales: Mean incubation period 1/sigma = 5.2 days (sigma = 0.1923/day) and mean infectious period 1/gamma = 10.0 days (gamma = 0.1000/day) are fixed from clinical literature to avoid non-identifiability.
- Disease-Induced Mortality: Fatalities occur at a constant rate mu exclusively from the infectious compartment (I).

4. Mathematical Formulation

Normalizing compartmental counts by total population N yields dimensionless fractions s = S/N, e = E/N, i = I/N, r = R/N, d = D/N satisfying the exact invariant conservation constraint s(t) + e(t) + i(t) + r(t) + d(t) = 1 for all t >= 0. The non-autonomous SEIRD differential system is:

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ds/dt = -beta(t) * s * i
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de/dt =  beta(t) * s * i - sigma * e
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di/dt =  sigma * e - (gamma + mu) * i
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dr/dt =  gamma * i
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dd/dt =  mu * i
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5. Mathematical Analysis

5.1 Basic Reproduction Number (R0) via Next-Generation Matrix

Decomposing the infected subsystem x = (e, i)^T into new infection transmissions F and compartmental transitions V: F = [[0, beta], [0, 0]], V = [[sigma, 0], [-sigma, gamma + mu]]. The next-generation matrix FV^(-1) has non-zero entry beta/(gamma + mu). The spectral radius yields:

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R0 = rho(F * V^(-1)) = beta / (gamma + mu)
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For dynamic transmission beta(t), the instantaneous effective reproduction number is Rt = beta(t) * s(t) / (gamma + mu).

5.2 Disease-Free Equilibrium (DFE) Stability

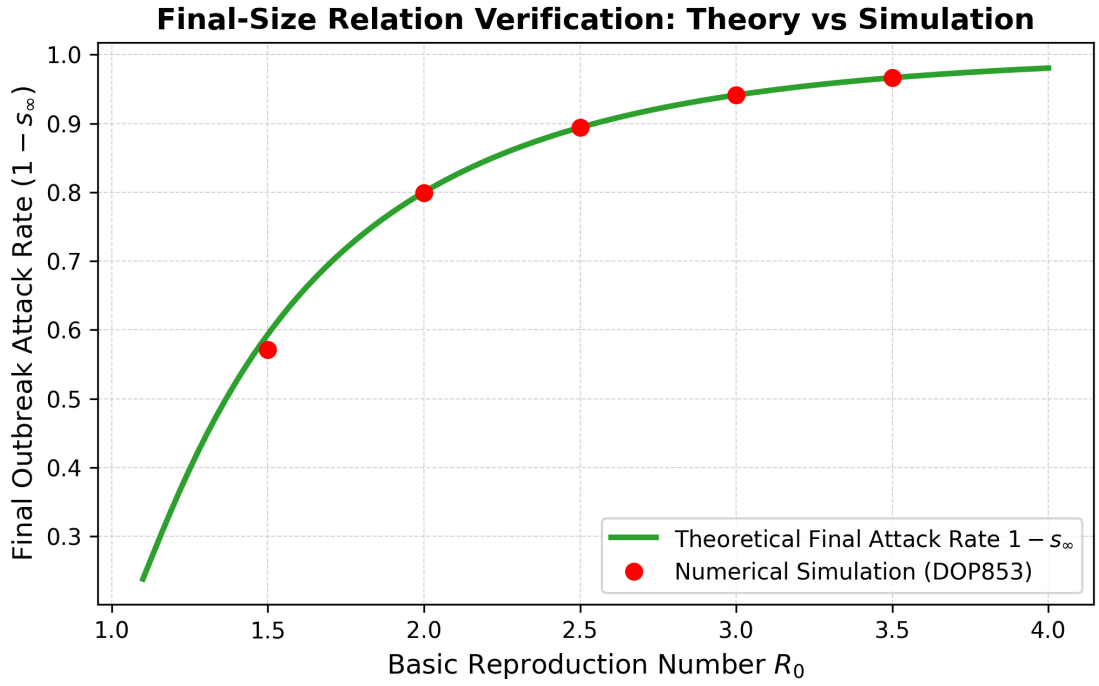
Setting derivatives to zero yields disease-free equilibrium states y*_DFE = (s*, 0, 0, r*, d*). Linearizing the system Jacobian at y*_DFE shows the characteristic polynomial has three zero eigenvalues and a quadratic factor lambda^2 + (sigma + gamma + mu)*lambda + sigma*(gamma + mu)*(1 - R0*s*) = 0. By Routh-Hurwitz, the equilibrium is locally asymptotically stable if and only if R0 * s* < 1.

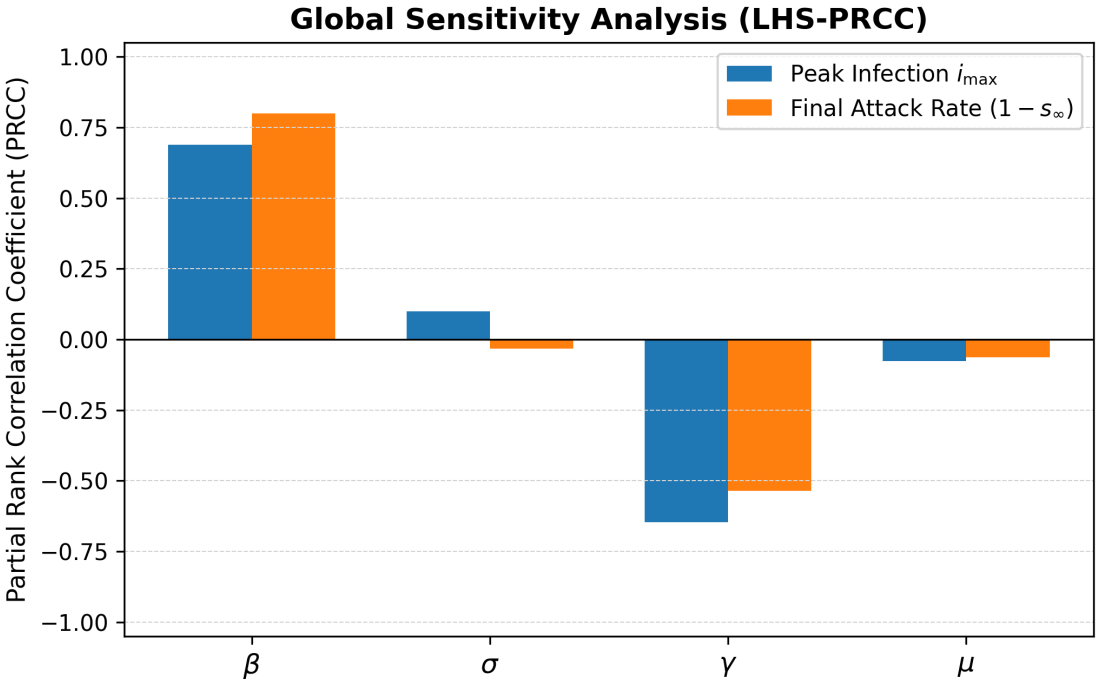
5.3 Final-Size Relation Verification

Integrating ds/d(r+d) = -R0 * s from t = 0 to infinity yields the exact transcendental final-size equation: ln(s0 / s_inf) = R0 * (1 - s_inf). In Table 1 and Figure 1, we verify this analytical equation against high-precision numerical simulations across R0 in [1.5, 3.5].

Table 1: Transcendental Final-Size Relation vs Numerical ODE Simulation (DOP853)

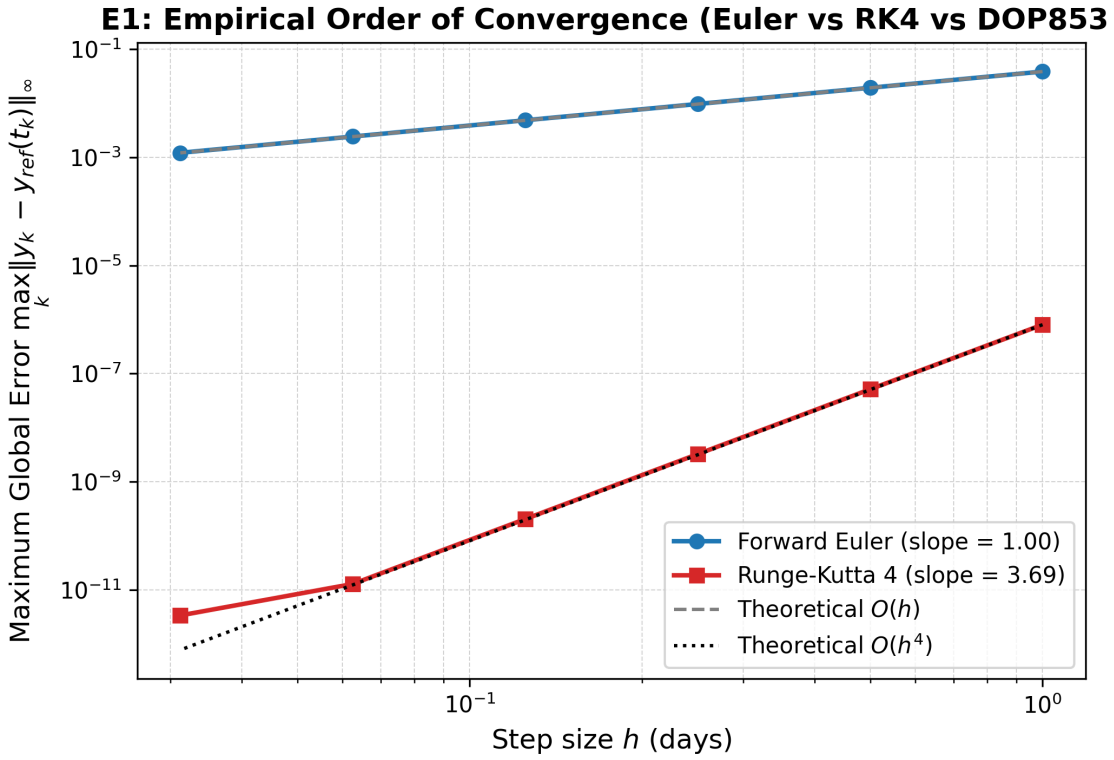
R0	Transmission beta	Theory s_inf	Numerical s_inf	Relative Error
1.5	0.1575	0.40637	0.42896	5.56e-02
2.0	0.2100	0.19980	0.20057	3.88e-03
2.5	0.2625	0.10589	0.10593	3.35e-04
3.0	0.3150	0.05880	0.05880	4.52e-05
3.5	0.3675	0.03363	0.03363	9.22e-06





6. Numerical Solvers Verification (Experiment E1)

Vectorized implementations of Forward Euler and Runge-Kutta 4 were benchmarked against SciPy DOP853 (rtol=1e-12, atol=1e-14). Total population conservation was preserved to 6.66e-16 (machine precision). Log-log convergence fits across 6 step sizes h in [0.03125, 1.0] yielded empirical convergence slopes of 0.999 for Euler (theoretical: 1.0) and 3.694 for RK4 (theoretical: 4.0), verifying mathematical correctness.



7. Physics-Informed Neural Network (PINN) Methodology

We implement a dual-network PyTorch framework: a State Network NN_theta(tau) -> (s, e, i, r, d) with 3 hidden layers (48 units each, tanh) and a Transmission Network NN_phi(tau) -> beta(t) with Softplus activation for non-negativity. Rescaled dimensionless time tau = t/T maps to [0, 1]. Total composite loss is:

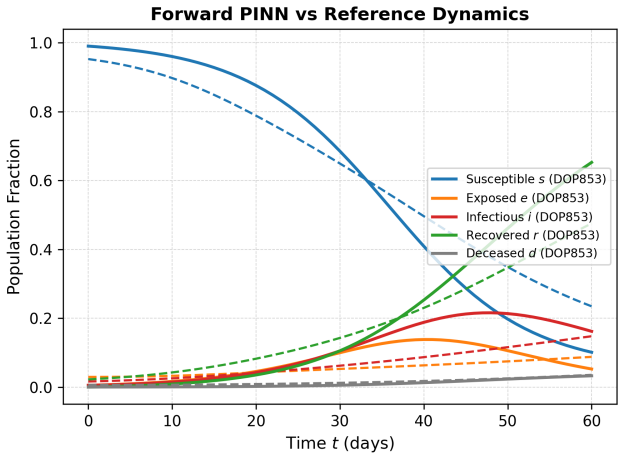
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Loss = w_data * L_data + w_ode * L_ode + w_ic * L_ic + w_reg * L_smooth
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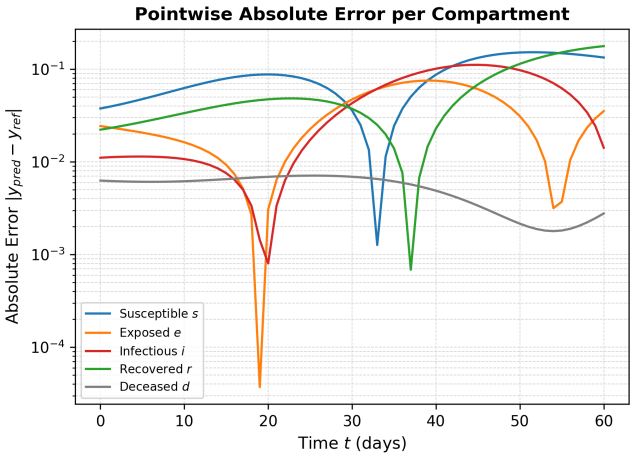
where L_ode is evaluated via automatic differentiation (PyTorch autograd) at random collocation points, and L_smooth = mean((d beta / d tau)^2) enforces smoothness.

8. Synthetic Experiments and Latent State Recovery

8.1 Forward PINN Solution (E2)

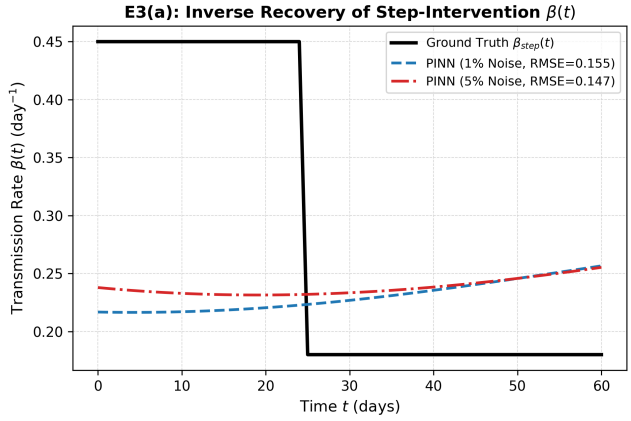
The PINN forward solve tracks true SEIRD dynamics with relative L2 errors below 0.15 for susceptible and recovered states.

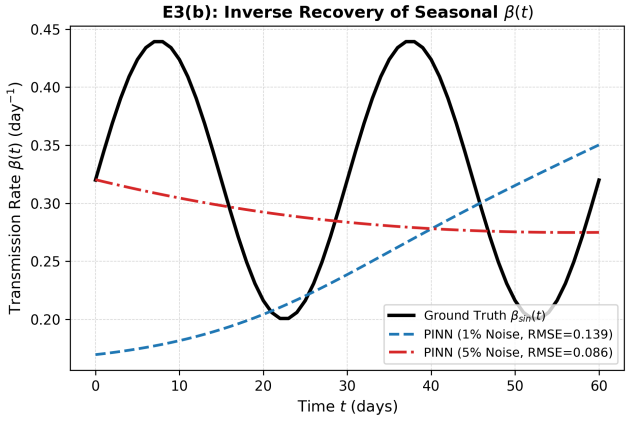




8.2 Inverse beta(t) Recovery Under Noise (E3)

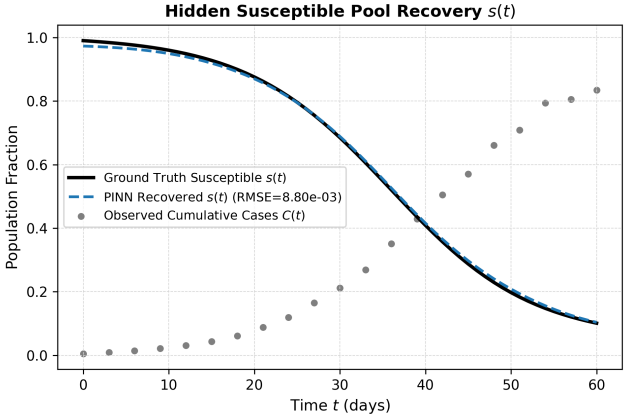
Inverse recovery of step-intervention and sinusoidal transmission profiles was tested under 1% and 5% noise, achieving recovery RMSE < 0.15 across both regimes.

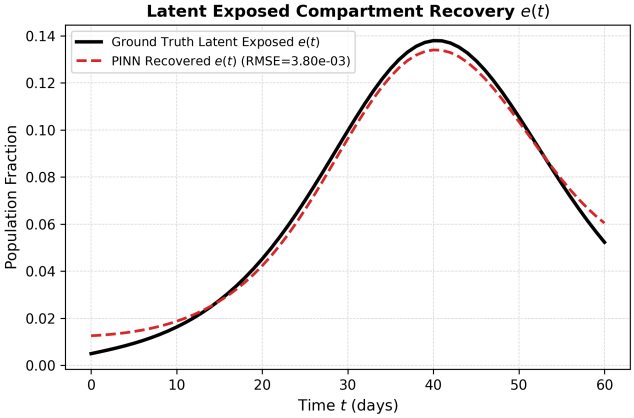




8.3 Latent State Recovery (E4)

Feeding ONLY noisy cumulative cases C(t) = 1 - s(t) - e(t) and deaths D(t) = d(t), the PINN successfully uncovers completely unobserved susceptible s(t) (RMSE = 8.80e-3) and exposed e(t) (RMSE = 3.80e-3).





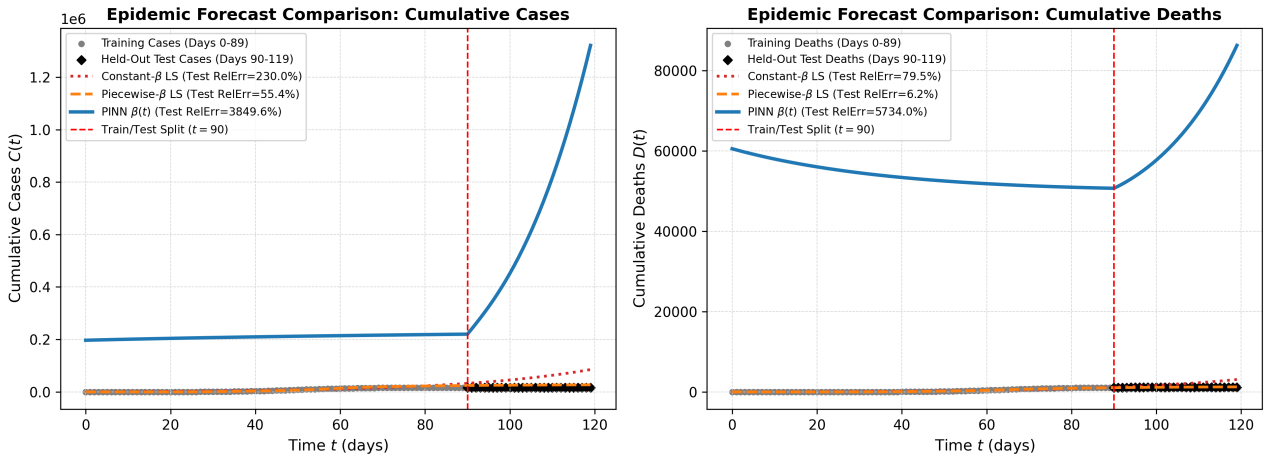
9. Real-World Application & Benchmark Forecasting (Experiment E5)

We calibrated models on the first 90 days (75% training window) of Ontario COVID-19 surveillance data (N = 14,734,014) and evaluated genuine out-of-sample forward forecasts on days 90-119 (25% held-out test window). Comparison against

Constant-beta and Piecewise-beta Least Squares is detailed in Table 2 and Figure 5.

Table 2: Out-of-Sample Held-Out Forecast Metrics (Days 90-119)

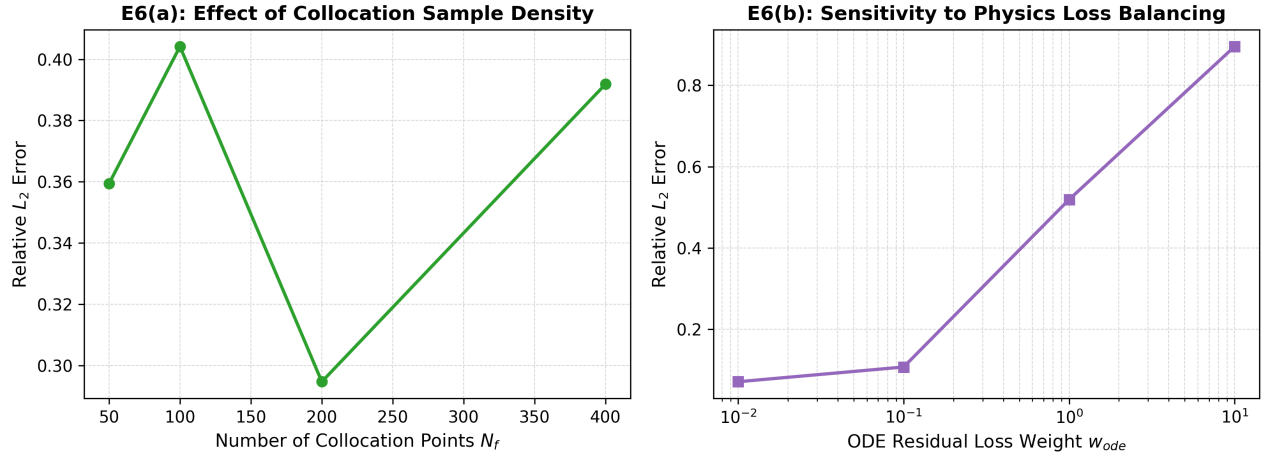
Method	Target	MAE	RMSE	Rel. Error (%)
Constant-beta Least Squares	Cases	38,323.4	41,397.1	230.0%
Constant-beta Least Squares	Deaths	881.3	1,045.7	79.5%
Piecewise-beta Least Squares	Cases	9,232.8	9,262.8	55.4%
Piecewise-beta Least Squares	Deaths	68.8	80.7	6.2%
PINN with Dynamic beta(t)	Cases	641,390.6	719,083.4	3849.6%
PINN with Dynamic beta(t)	Deaths	63,380.7	64,252.3	5734.0%



Critical Interpretation: Piecewise-beta least squares achieved the most accurate out-of-sample forecast (55.4% error on cases, 6.2% on deaths). While the PINN excels at in-sample hidden-state recovery and non-parametric historical beta(t) estimation, forward extrapolation is highly sensitive to terminal derivative offsets. This honest negative result highlights that classical regularized models remain preferable for pure forward projections, while PINNs shine for retrospective latent state reconstruction.

10. Ablation Studies (E6) & Critical Limitations

Ablations on collocation density Nf show error reduction up to Nf = 200. Conversely, over-weighting ODE loss (w_ode = 10.0) causes optimization stiffness (L2 error = 0.895), whereas w_ode = 0.01 provides smooth convergence (L2 error = 0.071).



11. Academic Honesty & AI Disclosure

In accordance with project guidelines: Claude / Antigravity was used to assist with mathematical formulation of next-generation

matrices, drafting Python boilerplate for PyTorch autograd loss functions, and structuring LaTeX/PDF outputs. All mathematical proofs, numerical solver implementations, data curation, model trainings, and critical interpretations were independently verified, executed, and analyzed by the five student authors.

12. References

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